1 **ACSBR/2025/I/10**

(a) Given that θ is a reflex angle and $\cos\theta = -\frac{3}{5}$, find the exact value of

(i) $\tan 2\theta$, [2]

(ii) $\operatorname{cosec}\theta$, [1]

(iii) $\sin(-\theta)$. [1]

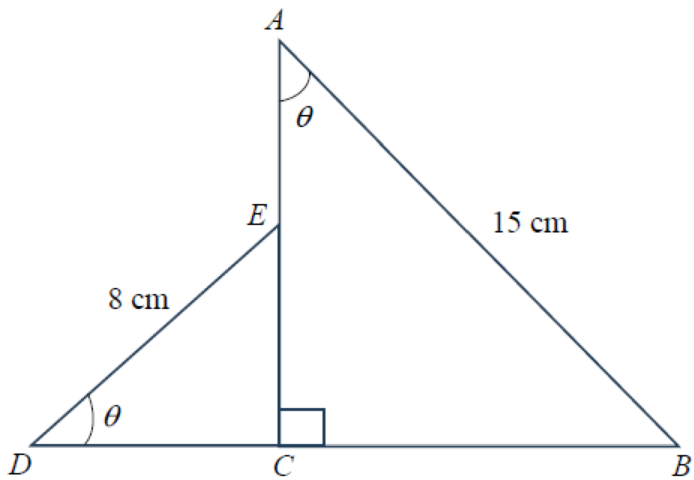
(b) State the values between which each of the following must lie:

(i) the principal value of $\sin^{-1} x$, [1]

(ii) the principal value of $\cos^{-1} x$. [1]

(c) **Without using a calculator**, show that $\sin 75° = \frac{\sqrt{6}+\sqrt{2}}{4}$. [3]

2 **ACSBR/2025/II/07**



(a) Show that the perimeter, P cm, of the figure $ABCDE$ can be expressed as $23\cos\theta + 7\sin\theta + 23$. [4]

(b) By expressing P in the form $R\cos(\theta - \alpha) + k$, where $R > 0$ and α is acute, find the maximum possible value of P and the corresponding value of θ. [6]

(c) Find the value of θ for which $P = 40$. [2]

3 **BASS/2025/II/01**

A calculator must not be used in this question.

(a) Show that $\sin 105° = \frac{\sqrt{2}+\sqrt{6}}{4}$, [4]

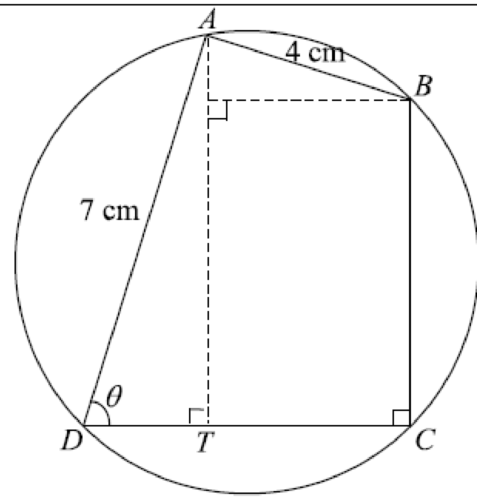
(b) Hence find the exact value of $\cos^2 105°$ in the form of $\frac{a-\sqrt{3}}{b}$, where a and b are integers. [4]

4 **BASS/2025/II/03**

(a) Show that $\dfrac{3\sin x\cos x - 2}{1-\sin^2 x} = 3\tan x - 2\sec^2 x$. [4]

(b) Hence find all the angles between 0 and 2π which satisfy $3\sin x\cos x - 2 = 7\sin^2 x - 7$. [4]

5 **BASS/2025/II/08**



The diagram shows a cyclic quadrilateral *ABCD*.

BD as the diameter of the circle and *AT* is perpendicular to *DC*.

$AB = 4$ cm, $AD = 7$ cm, angle $BCD = 90°$ and angle $ADC = \theta\,°$ where $0° < \theta < 90°$.

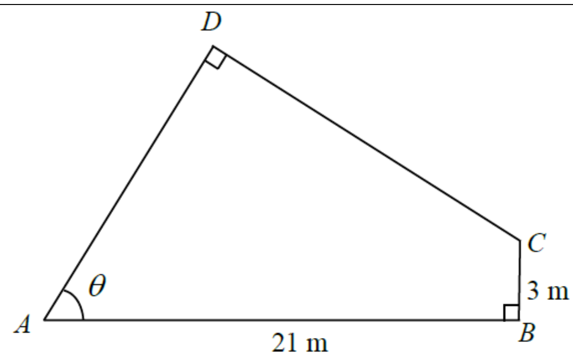
(a) Show that the perimeter of the quadrilateral *ABCD*, *P* cm, is given by $P = 11 + 3\cos\theta + 11\sin\theta$.

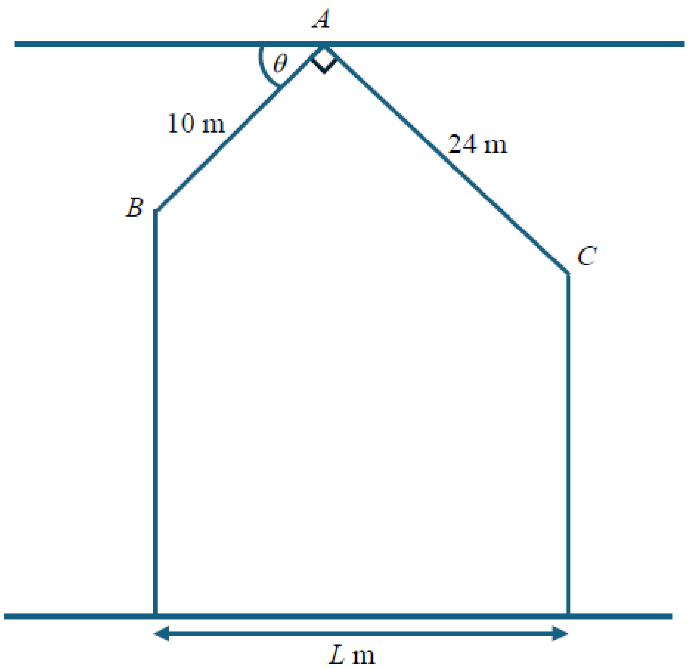
[4]

(b) Express *P* in the form $k + R\cos(\theta - \alpha)$, where $k > 0$, $R > 0$ and $0° < \alpha < 90°$. [3]

(c) Explain why the perimeter cannot be 24 cm. [1]

(d) Find the corresponding value of θ when the perimeter is at its maximum value [1]

6	**CCHY/2025/I/10**
	 The diagram shows a quadrilateral field *ABCD*, where $AB = 21$ m and $BC = 3$ m. Angle ABC = angle $ADC = 90°$. Angle $BAD = \theta$, for $0° < \theta < 90°$, and can vary. The perimeter of the fencing around the quadrilateral field *ABCD* is *P* m. **(i)** Show that $P = 24 + 18\cos\theta + 24\sin\theta$. [3] **(ii)** Express *P* in the form $R\sin(\theta + \alpha) + 24$, where $R > 0$ and $0° < \alpha < 90°$. [3] **(iii)** Given that the total perimeter of the fencing is 53 m, find the value(s) of θ. [2] **(iv)** Explain why the total length of the fencing will never exceed a certain value and state this value. [2]
7	**CCHY/2025/II/4**
	It is given that $f(x) = 3x^3 + ax^2 - 7x + 3$, where *a* is a constant, has a factor of $x + 1$. **(a)** Find the value of *a* and factorise $f(x)$ completely. [4] **(b)** Solve the equation $f(\cot^2\theta) = 0$ for $-90° < \theta < 90°$. [3]
8	**CCHY/2025/II/5**
	(a) Prove the identity $\sin 2x(a\cot x + b\tan x) = a + b + (a - b)\cos 2x$, where *a* and *b* are constants. [4] **(b)** Hence solve the equation $\sin\frac{1}{2}\theta\left(3\cot\frac{1}{4}\theta + 5\tan\frac{1}{4}\theta\right) = 7$ for $0 \le \theta \le 4\pi$. [3]

9	**GMS/2025/I/11**
	The diagram shows the cross-section of a house with a rooftop BAC. The length AB and AC are 10 m and 24 m respectively. The angle between AB and the horizontal through A is θ and angle $BAC = 90°$.  The base of the house is L m. **(a)** Show that $L = 24\sin\theta + 10\cos\theta$. [2] **(b)** Express L in the form $R\sin(\theta + \alpha)$, where $R > 0$ and α is an acute angle. [3] **(c)** Find the largest possible value of L and the corresponding value of θ. [2]

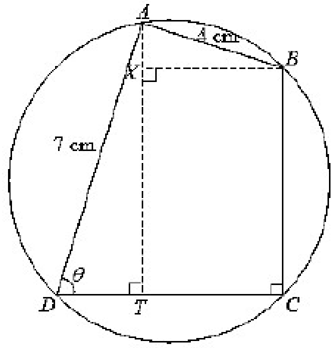
1	(a)(i)	$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$ $= \frac{2\left(\frac{4}{3}\right)}{1-\left(\frac{4}{3}\right)^2}$ $= \frac{\frac{8}{3}}{\frac{-7}{9}}$ $= -\frac{24}{7}$
	(a) (ii)	$\text{cosec}\,\theta = \frac{1}{\sin\theta}$ $= \frac{-5}{4}$
	(a)(iii)	$\sin(-\theta) = -\sin\theta = \frac{4}{5}$
	(b)(i)	$-\frac{\pi}{2} \le \sin^{-1} x \le \frac{\pi}{2}$ or $-90° \le \sin^{-1} x \le 90°$
	b(ii)	$0 \le \cos^{-1} x \le \pi$ or $0 \le \cos^{-1} x \le 180°$
	(c)	$\sin 75° = \sin(30° + 45°)$ $= \sin 30° \cos 45° + \cos 30° \sin 45°$ $= \frac{1}{2}\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\frac{\sqrt{2}}{2}$ $= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}$ $= \frac{\sqrt{6}+\sqrt{2}}{4}$
2	(a)	$DC = 8\cos\theta$ $CB = 15\sin\theta$ $EC = 8\sin\theta$ $AC = 15\cos\theta$ $P = 15 + 15\sin\theta + 8\cos\theta + 8 + 15\cos\theta - 8\sin\theta$ $= 23\cos\theta + 7\sin\theta + 23$
	(b)	$\tan\alpha = \frac{7}{23}$ $\alpha = 16.9275$ $= 16.9°$ $R = \sqrt{23^2 + 7^2} = \sqrt{578} = 24.0$ $P = 24\cos(\theta - 16.9°) + 23$ $P_{\max} = 24.0 + 23 = 47.0$ $\theta - 16.9° = 0$ $\theta = 16.9°$

3	**a**	$\sin 105° = \sin(60° + 45°)$ $= \sin 60° \cos 45° + \cos 60° \sin 45°$ $= \frac{\sqrt{3}}{2}\frac{\sqrt{2}}{2} + \frac{1}{2}\frac{\sqrt{2}}{2}$ $= \frac{\sqrt{2}+\sqrt{6}}{4}$ (Shown)
	b	$\cos^2 105° = 1 - \sin^2 105°$ $= 1 - \left(\frac{\sqrt{2}+\sqrt{6}}{4}\right)^2$ $= 1 - \left(\frac{2+2\sqrt{12}+6}{16}\right)$ $= \frac{8-4\sqrt{3}}{16}$ $= \frac{2-\sqrt{3}}{4}$
4	**a** **b**	$\text{LHS} = \frac{3\sin x \cos x - 2}{1-\sin^2 x}$ $= \frac{3\sin x \cos x - 2}{\cos^2 x}$ $= \frac{3\sin x \cos x}{\cos^2 x} - \frac{2}{\cos^2 x}$ $= \frac{3\sin x}{\cos x} - 2\sec^2 x$ $= 3\tan x - 2\sec^2 x$ $= \text{RHS}$ $3\sin x \cos x - 2 = 7\sin^2 x - 7$ $3\sin x \cos x - 2 = -7(1-\sin^2 x)$ $\frac{3\sin x \cos x - 2}{1-\sin^2 x} = -7$ $3\tan x - 2\sec^2 x = -7$ $3\tan x - 2(1+\tan^2 x) = -7$ $2\tan^2 x - 3\tan x - 5 = 0$ $(2\tan x - 5)(\tan x + 1) = 0$ $\tan x = \frac{5}{2}$ and $\tan x = -1$ $x = 1.1902, 4.3318826$; $x = 2.356194, 5.49778$ $\approx 1.19, 4.33$; $\approx 2.36, 5.50$

5

a

$$\frac{DT}{7} = \cos\theta$$
$$DT = 7\cos\theta$$



$$\frac{AT}{7} = \sin\theta$$
$$AT = 7\sin\theta$$

$\because BD = \text{diameter}$

$\therefore \angle BAD = 90°$

$\therefore \angle BAX = \theta°$

$$\frac{BX}{4} = \sin\theta$$
$$TC = BX = 4\sin\theta$$

$$\frac{AX}{4} = \cos\theta$$
$$AX = 4\cos\theta$$

$$P = 7 + 4 + (7\sin\theta - 4\cos\theta) + (4\sin\theta + 7\cos\theta)$$
$$= 11 + 3\cos\theta + 11\sin\theta \text{ (shown)}$$

b

$$R = \sqrt{3^2 + 11^2} = \sqrt{130}$$
$$\tan\alpha = \frac{11}{3}$$
$$\alpha = 74.745 \text{ (5sf)}$$
$$11 + 3\cos\theta + 11\sin\theta = 11 + \sqrt{130}\cos(\theta - 74.7) \text{ (1dp)}$$

c

$$\because -\sqrt{130} \le \sqrt{130}\cos(\theta - 74.7) \le \sqrt{130}$$
$$\therefore 11 - \sqrt{130} \le 11 + \sqrt{130}\cos(\theta - 74.7) \le 11 + \sqrt{130}$$
$$\therefore \max(11 + \sqrt{130}\cos(\theta - 74.7)) \le 11 + \sqrt{130}$$
$$\therefore P < 22.401...$$
$$\therefore P \ne 24$$

d

Corresponding $\theta - 74.7 = 0°$

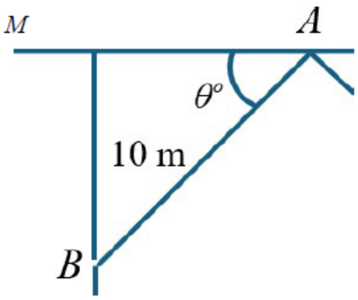
$\theta = 74.7°$ (1dp)

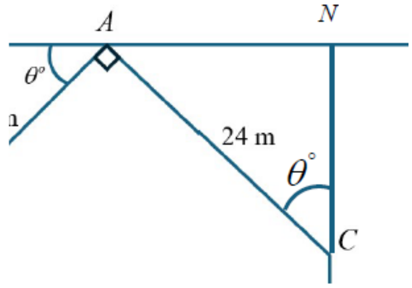
6	**(i)** $\sin\theta = \dfrac{BF}{21}$ $BF = 21\sin\theta$ $\cos\theta = \dfrac{EB}{3}$ $EB = 3\cos\theta$ $DC = 21\sin\theta - 3\cos\theta$ $\cos\theta = \dfrac{AF}{21}$ $AF = 21\cos\theta$ $\sin\theta = \dfrac{EC}{3}$ $EC = 3\sin\theta$ $AD = 21\cos\theta + 3\sin\theta$ Hence, $P = 21 + 3 + 21\sin\theta - 3\cos\theta + 21\cos\theta + 3\sin\theta$ $= 24 + 18\cos\theta + 24\sin\theta$ (shown) **(ii)** $R = \sqrt{18^2 + 24^2}$ $= \sqrt{900}$ $= 30$ $\tan\alpha = \dfrac{18}{24}$ $= 36.8699°$ $= 36.9°$ $\therefore P = 30\sin(\theta + 36.9°) + 24$ **(iii)** $30\sin(\theta + 36.8699°) + 24 = 53$ $\sin(\theta + 36.8699°) = \dfrac{29}{30}$ Basic angle $= \sin^{-1}\left(\dfrac{29}{30}\right)$ $= 75.1649°$ $\theta + 36.8699° = 75.1649°, 104.8351°$ $\theta = 38.3°,\ 68.0°$ **(iv)** $-1 \le \sin(\theta + 36.9°) \le 1$ $-30 \le 30\sin(\theta + 36.9°) \le 30$ $-6 \le 30\sin(\theta + 36.9°) + 24 \le 54$ The total length will not exceed 54m as the max length is 54m.

7	**(a)** $\mathrm{f}(-1)=3(-1)^3+a(-1)^2-7(-1)+3$ $0=-3+a+7+3$ $a=-7$ By long division, $\mathrm{f}(x)=3x^3-7x^2-7x+3$ $=(x+1)(3x^2-10x+3)$ $=(x+1)(3x-1)(x-3)$ **(b)** $\cot^2\theta=-1$ (rej) or $\cot^2\theta=\frac{1}{3}$ or $\cot^2\theta=3$ $\cot\theta=\pm\sqrt{3}$ $\cot\theta=\pm\frac{1}{\sqrt{3}}$ $\theta=\pm 30°$ $\theta=\pm 60°$
8	**(a)** $\sin 2x(a\cot x+b\tan x)=2\sin x\cos x\left(\frac{a\cos x}{\sin x}+\frac{b\sin x}{\cos x}\right)$ $=2a\cos^2 x+2b\sin^2 x$ $=a(\cos 2x+1)+b(1-\cos 2x)$ $=a\cos 2x+a+b-b\cos 2x$ $=a+b+(a-b)\cos 2x$ **(b)** $3+5+(3-5)\cos 2\left(\frac{1}{4}\theta\right)=7$ $-2\cos\left(\frac{1}{2}\theta\right)=-1$ $\cos\left(\frac{1}{2}\theta\right)=\frac{1}{2}$ basic angle $=\cos^{-1}\frac{1}{2}=\frac{\pi}{3}$ $\frac{1}{2}\theta$ is in 1st or 4th quadrant. $\frac{1}{2}\theta=\frac{\pi}{3}$ or $2\pi-\frac{\pi}{3}$ $\theta=\frac{2\pi}{3}$ or $\frac{10\pi}{3}$

9

(a)





$\cos\theta = \dfrac{AM}{10}$

$AM = 10\cos\theta$

$\sin\theta = \dfrac{AN}{24}$

$AN = 24\sin\theta$

$L = AN + AM = 24\sin\theta + 10\cos\theta$ (Shown)

Candidates must show a degree of proving from trigo identities

(b)

$R = \sqrt{10^2 + 24^2} = 26$

$\alpha = \tan^{-1}\left(\dfrac{10}{24}\right) = 22.620^o$

$L = 26\sin(\theta + 22.6)$

(c)

Largest possible L = 26 m

$\sin(\theta + 22.6^o) = 1$

$\theta + 22.6^o = 90^o$

$\theta = 67.4^o$